Module6-Object Analysis Classification

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- ✓Classification Theory
- ✓Approaches for Identifying Classes-Noun Phrase Approach
- ✓Selecting classes from the relevant and fuzzy categories
- ✓Common Class Pattern
- ✓Use Case Driven Approach
- ✓Class Responsibility Collaborator(CRC) Approach.

Object Analysis: Classification

✓The concept of classification

✓Identification of classes with:

- **Noun phrase approach**
- **Common class patterns approach**
- **Sequence/collaboration modeling**
- **Classes, Responsibilities, and Collaborators (CRC) approach**

Object Analysis: Classification

Introduction

✓**Object –oriented analysis** is a process by which we can identify classes that play a role in achieving system goals and requirements.

✓Unfortunately, classes rarely are there for the picking.

✓**Identification of classes is the hardest part of object-oriented analysis.**

Object Analysis: Classification

✓**Identification of classes**

What did they say:

Booch: "There is no such a thing as the perfect class structure, nor the right set of objects .

As in any engineering discipline, our design choice is compromisingly shaped by many competing factors"

Gaberial, et al. (the designers of OO language CLOS): "That is a fundamental question for which there is no easy answer. I try things".

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Object Analysis: Classifications Theory

Classification: the process of checking to see if an object belongs to a category or class.

- ✓Classification is regarded as a basic attribute of human nature.
- ✓Human beings classify information every instant of their waking lives.
- ✓A human being is a very sophisticated information system, partly he or she posses a superior classification capability.
- ✓**For example: Identification of car- clearly you have a general idea of cars look like, sound like, do and are good for-you have a notion of car-kind or in object-oriented terms, the class car.**

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Object Analysis: Classifications Theory

✓**Object-Oriented Analysis and Design (OOAD) uses classification** as a fundamental concept to organize and model the real-world domain into classes and objects. Here are some examples:

1. **Animal Classification**:

   Classes: Mammal, Bird, Fish.

   Objects: Dog (Mammal), Sparrow (Bird), Salmon (Fish).

   Mammals might have attributes like "warm-blooded" and "has fur," while Birds could have "feathers" and "can fly." Behavior like "movement" or "feeding" can be shared across classes but implemented differently.

2. **Library System:**

   Classes: Book, Member, Librarian.

   Objects: "Introduction to OOAD" (Book), "Gunasekaran" (Member), "Library Assistant #3" (Librarian).

   Books could have attributes like "title," "author," and "ISBN," while Members could have "membership ID" and "borrowedBooks." The "borrowBook()" behavior involves interactions among these objects.

Object Analysis: Classifications Theory

3. School Management:

- Classes: Student, Teacher, Classroom.
- Objects: "John Doe" (Student), "Mrs. Sharma" (Teacher), "Room 101" (Classroom).
- Students might have attributes such as "grade" and "attendance," while Teachers might have "subject expertise" and "salary." Interactions like "assignHomework()" or "markAttendance()" demonstrate behaviors.

Classification in OOAD helps to create a logical structure, making the system easier to design, understand, and maintain.

Object Analysis: Classifications Theory-(continued)

Classes are important mechanisms for classifying objects.

- ✓The chief role of class is **to define the attributes, methods, and applicability of its instances**.
- ✓The class car, for example, defines the property color. Each individual car-formally each instance of the class car- will have a property, such as maroon, yellow or white.
- ✓It is fairly natural to partition the world into objects that have properties (attributes) and methods (behaviors).
- ✓It is common and useful partitioning or classification, but we also routinely divide the world along a second dimension: **We distinguish classes from instances**.

CLASS is specification of structure, behavior, and the description of an object-the conceptual building block for designing systems.

CLASSIFICATION is more concerned with identifying the class of an object than the individual objects within a system.

Object Analysis: Classifications Theory

- ✓Tom and Gonzalez describe the recognition of concrete patterns or classes by humans as a psycho-physiological problem that involves a relationship between a person and a physical stimulus.
- ✓When you perceive a real-world object, you make an inductive inference and associate this perception with some general concepts or clues that you have derived from your past experience.
- ✓Human recognition, in reality is a question of estimating the relative odds that the input data can be associated with a class from a set of known classes, which depend on our past experiences and clues for recognition.
- ✓Martin and Odell observe, that an object can be categorized in more than one way. For example: a person may classify the object Betty as Woman. Her boss regards as an ***employee***. The person who mows her lawn calls her as ***Employer.*** The local animal control agency licenses her as a ***Pet Owner.***

Object Analysis: Classifications Theory

✓The problem of classification can be regarded as one discriminating things, not between the individual objects but between classes, via the search for features or invariant attributes or behaviors among members of a class.

✓**Classification: It can be defined as the categorization of input data (things) into identifiable classes via the extraction of significant features of attributes of the data from a background of irrelevant detail.**

Approaches for identifying classes

✓**Noun phrase approach.**
✓**Common class patterns approach.**
✓**The use-case driven, Sequence/collaboration modeling**
✓**Classes, Responsibilities, and Collaborators (CRC) approach.**

Approaches for identifying classes- Noun phrase approach

- ✓It was proposed by **Rebecca Wirfs-Brock, Brain Wilkerson and Lauren Wiener.**
- ✓In this method you go through requirements or use cases looking for noun phrases.
- ✓**Nouns in the textual description are considered to be classes and verbs to be methods of the classes.**
- ✓**All plurals are changed to singular, the nouns are listed, and the list divided into three categories: relevant classes, fuzzy classes, and irrelevant classes.**
- ✓It is safe **to scrap irrelevant classes**, **which either have no purpose or will be unnecessary.**
- ✓**Candidate classes are selected from other two categories.**
- ✓You must be able to **formulate a statement of purpose** or **simply eliminate it.**

Approaches for identifying classes- Noun phrase approach

Identifying Tentative classes

- ✓Look for nouns and noun phrases in the use cases
- ✓Some classes are implicit or taken from general knowledge.
- ✓All classes must make sense in the application domain; avoid computer implementation classes-defer them to design stage.
- ✓Carefully choose and define class name.

Approaches for identifying classes- Noun phrase approach

Selecting classes from the relevant and fuzzy categories

✓ **The following guidelines help in selecting candidate classes from the relevant and fuzzy categories of classes in the problem domain.**

✓ **Redundant classes:**
- Do not keep two classes that express the same information
- If more than one word is being used to describe the same idea, select the one that is the most meaningful in the context of the system
- (This is part of building a common vocabulary for the system as a whole. Choose your vocabulary carefully; use the word that is being used by the user of the system).

✓ **Adjective classes:**
- **Wirfs-Brock, Wilkerson and Weiner warn us about adjectives:** "Be wary of the use of adjectives. Adjectives can be used in many ways. An adjective can suggest a different kind of object, different use of the same object, or it could be utterly irrelevant. Does the object represented by the noun behave differently when the adjective is applied to it? If so make a new class.
- **For eg. Adult members behave differently than youth members, so the two could be classified into different classes.**

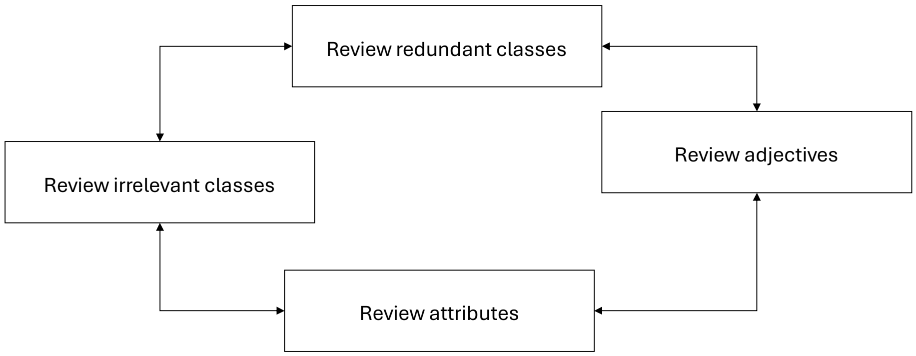
Approaches for identifying classes- Noun phrase approach

Selecting classes from the relevant and fuzzy categories

- **Attribute classes:**
  - Tentative objects that are used only as values should be defined or restated as attributes and not as a class.
  - **For eg. Client Status and Demographic of Client are not classes but attributes of the Client class.**
- **Irrelevant(fuzzy) classes:**
  - Each class must have a purpose, and every class should be clearly defined and necessary.
  - You must formulate a statement of purpose for each candidate class.
  - **If you cannot come up with a statement of purpose, simply eliminate the candidate class.**

  Note: Like any other activity in software development process, identifying relevant classes and eliminating irrelevant classes is an incremental process.

The Process of eliminating the redundant classes and refining the remaining classes is not sequential. You can move back and forth among these steps as often as you like.



ViaNet Bank ATM System-identifying classes by using Noun Phrase approach

✓The bank client must be able to deposit an amount to and withdraw an amount from his or her accounts using the touch screen at the ViaNet bank ATM kiosk. Each transaction must be recorded, and all the clients must be able to review all transactions performed against a given account. Recorded transactions must include the date, time, transaction type, amount, and account balance after the transactions.

- A ViaNet bank client can have two types of accounts: a savings account and a current account. For each current account, one related savings account can exist.
- Access to the ViaNet Bank accounts is provided by a PIN code consisting of four integer digits between 0 and 9.
- One PIN code allows access to all accounts held by a bank client.
- No receipts will be provided for any account transactions.
- The bank application operates for a single banking institution only.
- Neither a current nor savings account can have a negative balance. The system should automatically withdraw money from a related savings account if the requested withdrawal amount on the checking account is more than its current balance. If the balance on a savings account is less than the withdrawal amount requested, the transaction will stop, and the bank client will be notified.

ViaNet Bank ATM System-identifying classes by using Noun Phrase approach

Initial list of Noun Phrases: Candidate classes

- **Account** **Account Balance** **Amount**
- **Approval Process** **ATM Card** **ATM Machine**
- **Bank** **Bank Client** **Card**
- **Cash** **Check** **Checking**
- **Checking account** **Client** **Client's account**
- **Currency** **Dollar** **Envelope**
- **Four Digits** **Fund** **Invalid Pin**
- **Message** **Money** **Password**
- **PIN** **PIN Code** **Record**
- **Savings** **Savings Account** **Step**
- **System** **Transaction** **Transaction History**

ViaNet Bank ATM System-identifying classes by using Noun Phrase approach

Irrelevant classes

- Envelope
- Four Digits
- Step

ViaNet Bank ATM System-identifying classes by using Noun Phrase approach

List of Noun Phrases after canceling irrelevant classes

- Account Account Balance Amount
- Approval Process ATM Card ATM Machine
- Bank Bank Client Card
- Cash Check Checking
- Checking account Client Client's account
- Currency Dollar ~~Envelope~~
- ~~Four Digits~~ Fund Invalid Pin
- Message Money Password
- PIN PIN Code Record
- Savings Savings Account ~~Step~~
- System Transaction Transaction History

ViaNet Bank ATM System-identifying classes by using Noun Phrase approach

Reviewing the redundant Classes and Building a Common Vocabulary

- Redundant Classes Chosen Class
- Bank Client, Client = Bank Client
- Account, Client's account = Account
- Checking, Checking Account = Checking account
- PIN , PIN Code = PIN
- Savings, Savings Account = Savings Account
- Fund, Money = Fund
- ATM Card, Card = ATM Card

ViaNet Bank ATM System-identifying classes by using Noun Phrase approach

Reviewing the redundant Classes and Building a Common Vocabulary-Revised list

• Account	Account Balance	Amount
• Approval Process	ATM Card	ATM Machine
• Bank	Bank Client	~~Card~~
• Cash	Check	~~Checking~~
• Checking account	~~Client~~	~~Client’s account~~
• Currency	Dollar	~~Envelope~~
• ~~Four Digits~~	Fund	Invalid Pin
• Message	~~Money~~	Password
• PIN	~~PIN Code~~	Record
• ~~Savings~~	Savings Account	~~Step~~
• System	Transaction	Transaction History

ViaNet Bank ATM System-identifying classes by using Noun Phrase approach

Reviewing classes containing Adjectives

- The main question is:
  - ✓Does the object represented by the noun behave differently when the adjective is applied to it?
  - ✓If an adjective suggests a different kind of class or the class represented by the noun behaves differently when the adjective is applied to it, then we need to make a new class.
  - ✓However, if it is a different use of the same object or the class is irrelevant, we must eliminate it.
  - ✓In this example, we have no classes containing adjectives that we can eliminate.

ViaNet Bank ATM System-identifying classes by using Noun Phrase approach

Reviewing the possible attributes-identifying noun classes that are attributes and not classes

✓**Amount:** A value and not a class
✓**Account Balance:** an attribute of the Account class
✓**Invalid PIN:** It is only a value, not a class
✓**Password:** An attribute, possibly of the BankClient class.
✓**Transaction History:** An attribute possibly of the transaction class
✓**PIN:** An attribute of the BankClient class

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ViaNet Bank ATM System-identifying classes by using Noun Phrase approach

Reviewing the possible attributes-identifying noun classes that are attributes and not classes-Revised list

• **Account**	Account Balance	Amount
• Approval Process	**ATM Card**	**ATM Machine**
• **Bank**	**Bank Client**	Card
• Cash	Check	Checking
• **Checking account**	Client	Client's account
• Currency	Dollar	Envelope
• ~~Four Digits~~	Fund	Invalid Pin
• Message	Money	Password
• ~~PIN~~	~~PIN Code~~	Record
• Savings	**Savings Account**	Step
• ~~System~~	**Transaction**	~~Transaction History~~

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ViaNet Bank ATM System-identifying classes by using Noun Phrase approach

Reviewing the class purpose

✓Identifying the classes play a major role in achieving system goals and requirements is a major activity of OOA.

✓Each class must have a purpose.

✓Every class should be clearly defined and necessary in the context of achieving the system's goals.

✓If you cannot formulate a statement of purpose for a class, simply eliminate it.

✓If we delete the classes that add no purpose to the system, the remaining list will give the candidate classes of the system.

ViaNet Bank ATM System-identifying classes by using Noun Phrase approach

Reviewing the class purpose-List of candidate classes

✓**ATM Machine class:** Provides an interface to the Bank

✓**ATMCard Class:** Provides a client with a key to an account

✓**BankClient class:** A client is an individual that has a checking account and, possibly, a savings account

✓**Bank Class:** Bank clients belong to the Bank. It is a repository of accounts and processes the accounts' transactions.

✓**Account class:** An account class is a formal (abstract) class, it defines the common behaviors that can be inherited by more specific classes such as CheckingAccount and SavingsAccount.

✓**CheckingAccount class:** It models a clients checking account and provides more specialized withdrawal service.

✓**SavingsAccount class:** It models a client's savings account.

✓**Transaction class:** Keeps track of transaction, time, date, type, amount, and balance.

ViaNet Bank ATM System-identifying classes by using Noun Phrase approach

Final remarks

✓Identification of classes is an incremental process.

✓One can see that the list if candidate classes obtained is not complete. It needs to be refined further.

✓Unless you are starting with a lot of domain knowledge, you will miss more classes than you will eliminate.

✓**The major problem with the noun phrase approach is that it depends on the completeness and correctness of the available document, which is rare in real life.**

✓**On the other hand, large volumes of text on system documentation might lead to too many candidate classes.**

✓Even so, the noun phrase exercise can be very educational and useful if combined with other approaches, especially with the use cases as we did here.

ViaNet Bank ATM System-identifying classes by using COMMON CLASS PATTERNS(CCP) approach

✓The CCP approach is based a knowledge base of the common classes that have been proposed by various researchers such as **Shaler and Mellor, Ross and Coad and Yourdon.**

✓They have compiled and listed the following patterns for finding the candidate class and object:

- **Name: Concept Class**
- **Name: Event Class**
- **Name: Organization Class**
- **Name: People Class**
- **Name: Places Class**
- **Name: Tangible things and devices Class**

✓We shall now see the use context of these CCP.

ViaNet Bank ATM System-identifying classes by using COMMON CLASS PATTERNS approach

✓**Name: Concept Class**

- **Context:** A concept is a particular **idea or understanding** that we have of our world. It encompasses principles that are not tangible but used to organize or keep track business activities or communications.
- **Martin and Odell describe:** Privately held ideas or notions are called conceptions. When an understanding is shared by another, it becomes a concept.
- **Example: Performance is an example of concept class.**

✓**Name: Event Class**

- **Context:** Event classes are **points in time that must be recorded**. Things happen, usually to something else at a given date and time or as step in an ordered sequence.
- Associated with things remembered are attributes (after all things to remember are objects) **such as who, what, when, where, how, or why.**
- **Example: Landing, interrupt, request, and order.**

✓**Name: Organization Class**

- **Context:** An organization class is **a collection of people, resources, facilities or groups to which the users belong**; their capabilities have a defined mission, whose existence is largely independent of the individuals.
- **Example: An accounting department**

ViaNet Bank ATM System-identifying classes by using COMMON CLASS PATTERNS approach

✓**Name: People Class (also known as person, roles and roles played class)**

- **Context.** The people class represents **the different roles users play in interacting with the application.** People carry out some function. What role does a person play in the system?
- **Coad and Yourdon** explain that a class which is represented by a person can be divided into two parts:
  - **Those representing the users of the system** (such as an operator or clerk who interacts with the system; and
  - **Those representing people who do not use the system** but about whom information is kept by the system.
- **Example. Employee, client, teacher, and manager**

✓**Name: Places Class**

- **Context.** Places are **physical locations** that the system must keep information about.
- **Example. Buildings, stores, sites, and offices.**

✓**Name: Tangible things and devices Class**

- **Context.** This class includes **physical objects or groups of objects** that are tangible and devices with which the application interacts.
- **Example. Cars are example of tangible things, and pressure sensors are an example of devices**

ViaNet Bank ATM System-identifying classes by using COMMON CLASS PATTERNS approach

Identifying CCP

✓**Event classes:**
- Account class
- Checking Account class
- Savings Account class
- Transaction class

✓**Organization Classes:**
- Bank Class

✓**People and person classes:**
- Bank Client class

✓**Place classes:**
- Not applicable to this exercise

✓**Tangible things and devices class:**
- ATM Machine Class

Use-Case driven approach: identifying classes and their behaviors through scenario or collaboration modeling

✓**Use cases** are employed to model the scenarios in the system and specify **what external actors interact with the scenarios.**

✓Scenarios are described in text or through a sequence of steps.

✓**Use-Case modeling** is considered **a problem-driven approach to OOA**, in that the designer first considers the problem at hand and not the relationship between objects, as in data-driven approach.

✓**Modeling with use cases is a recommended aid in finding the objects** of a system and is the technique used by the unified approach.

✓Once the system is **described in terms of its scenarios**, the modeler can examine the **textual description** or steps of each scenario **to determine what objects are needed for the scenario to occur.**

✓**The process of creating sequence or collaboration diagrams is a systematic way to think about a use case (scenario) can take place; and by doing so, it forces you to think about objects involved in your application.**

Use-Case driven approach: identifying classes and their behaviors through scenario or collaboration modeling

- ✓Implementation of scenarios
- ✓The UML specification recommends that **at least one scenario for each significantly different use-case instance.**
- ✓Each scenario shows a different sequence of interaction between actors and the system, with all decisions definite. This process helps us to understand the behavior of system's objects.
- ✓**When you have reached the lowest use-case level,** you may create child sequence diagram or accompanying collaboration for the use-case. With the sequence and collaboration diagrams, you can model the implementation of the scenario.
- ✓Like the use-case diagrams, sequence diagrams are used to model scenarios in the systems.
  - **Use cases and the steps or textual description that define them offer a high-level view of a system.**
  - **Sequence diagram enables you to model a more specific analysis and also assists in the design of the system by modeling the interactions between objects in the system.**

Bank ATM system: Decomposing a Use-Case Scenario with a sequence diagram: Object Behavior Analysis

- ✓**Low level (executable) use cases of ATM system:**
  - Deposit Checking Deposit Savings
  - Invalid PIN Withdraw Checking
  - Withdraw more from Checking
  - Withdraw savings Withdraw savings denied
  - Checking Transaction History
  - Savings Transaction History
- ✓Let us create a sequence/collaboration diagram for the following use-cases:
  - **Invalid PIN use case**
  - **Withdraw Checking use case**
  - **Withdraw More from Checking use case**

Bank ATM system: Decomposing a Use-Case Scenario with a sequence diagram: Object Behavior Analysis

✓Creation of a sequence diagram for the Invalid PIN use case:
- **Insert the ATM Card**
- **Enter the PIN Number.**
- **Remove the ATM Card**

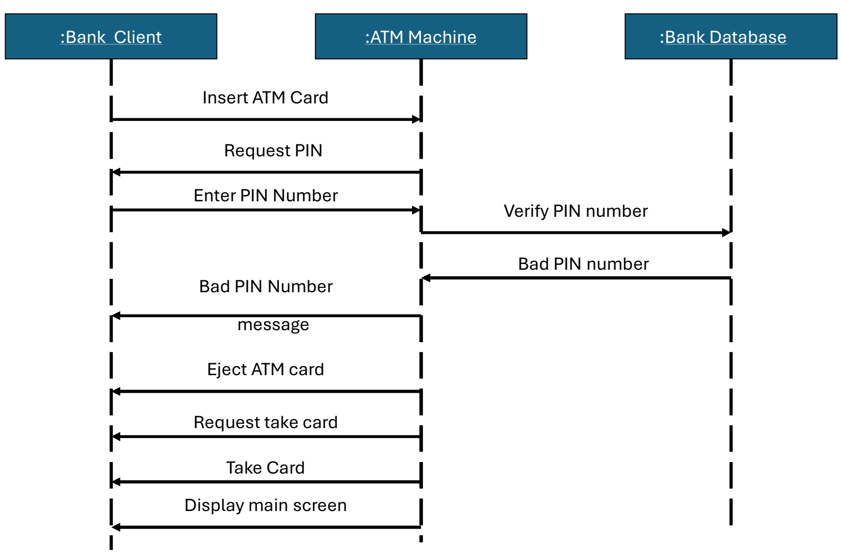
✓Based on these activities, the system should either grant the access right to the account or reject the card.

✓Next, we need to more explicitly define the system.

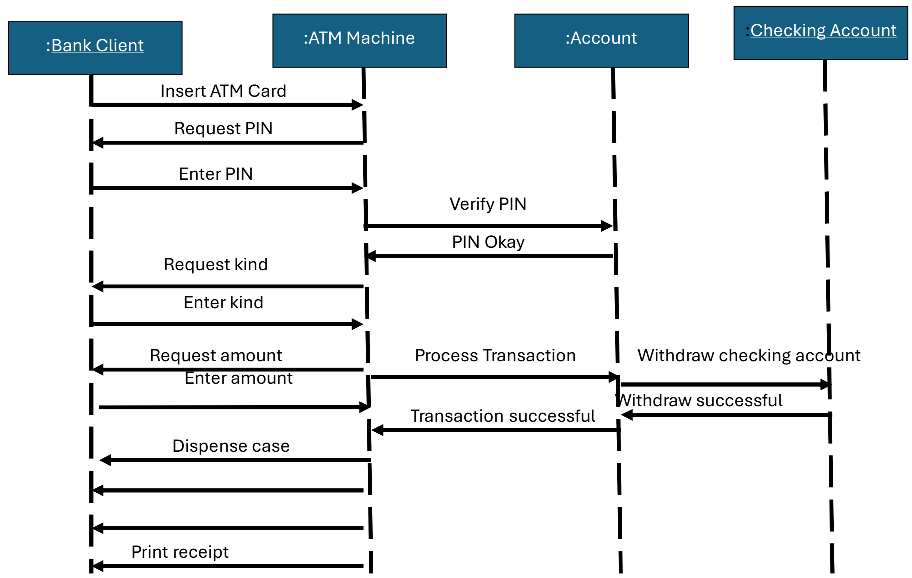
✓With what are we interacting?
- **We are interacting with an ATM Machine and the BankClient. So, the other objects of this use case are ATM Machine and Bank Database or Bank.**

✓Collaboration diagrams are just another view of sequence diagrams and therefore can be created automatically; most UML modeling tools automatically create them.

Sequence diagram for the Invalid PIN use case



Sequence diagram for the Withdraw Checking use case



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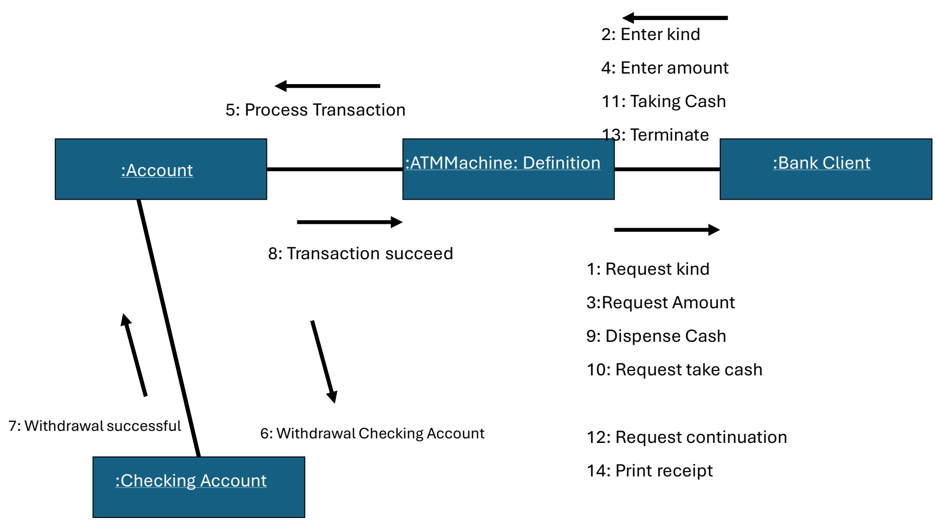
The collaboration diagram for the withdraw checking use case



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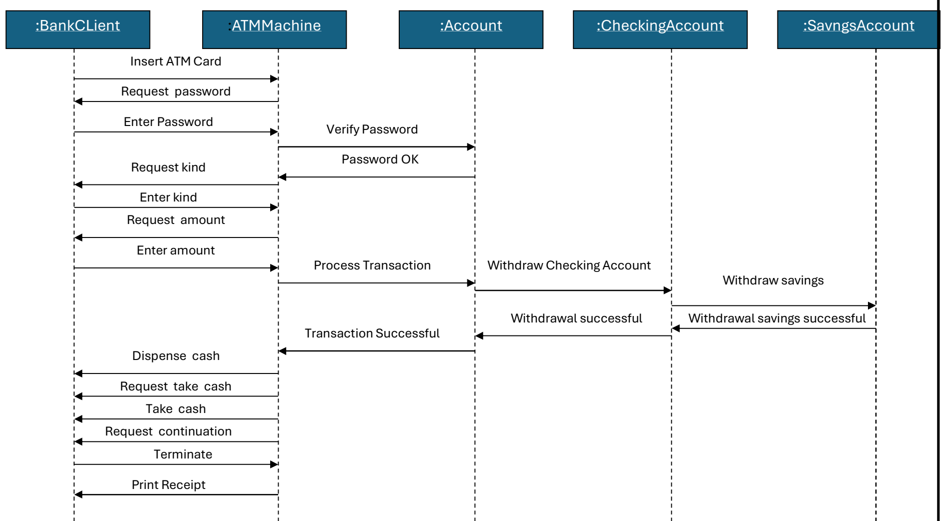
Sequence Diagram for the withdraw more from checking use case



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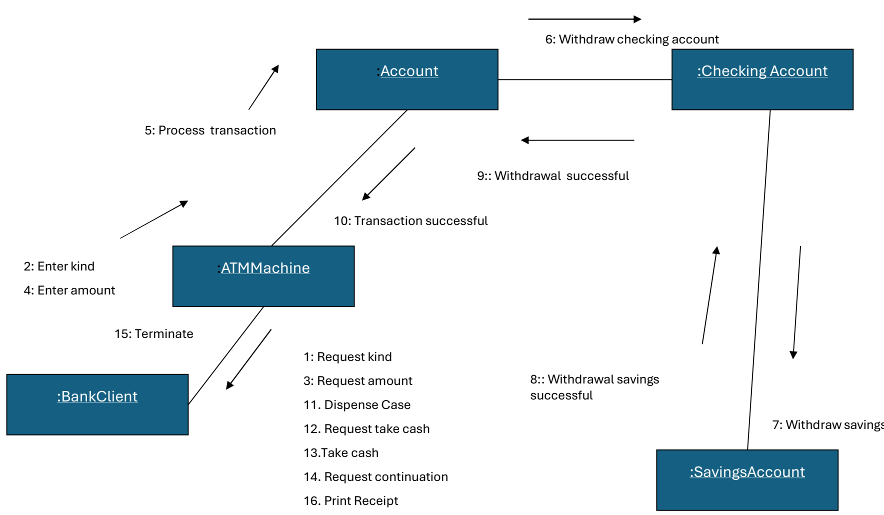
Collaboration Diagram for the withdraw more from checking use case



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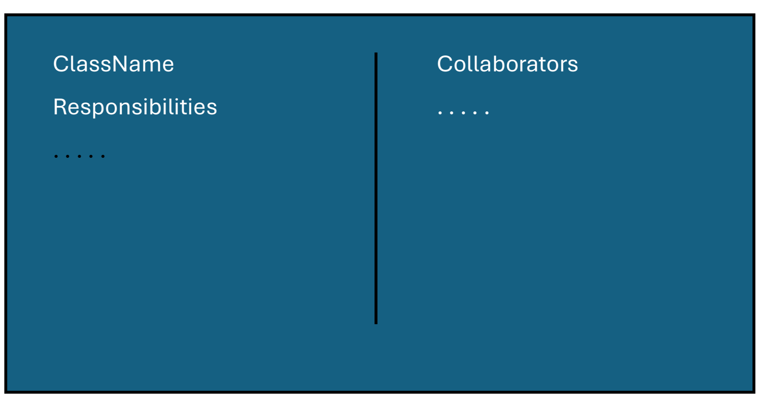
CLASSES, RESPONSIBILITIES, AND COLLABORATORS (CRC)

- ✓Developed by Cunningham, Wilkerson, and Beck.
- ✓**CRC is a technique used to identifying classes' responsibilities and therefore their attributes and methods**.
- ✓CRC can help us in identify classes.
- ✓CRC is a teaching technique than a method for identifying classes.
- ✓CRC is based on the idea that an object either can accomplish a certain responsibility itself, or it may require the assistance of other objects.
- ✓If it requires the assistance of other objects, it must collaborate with those objects to fulfill its responsibility.
- ✓CRC cards are 4"x6"index cards.

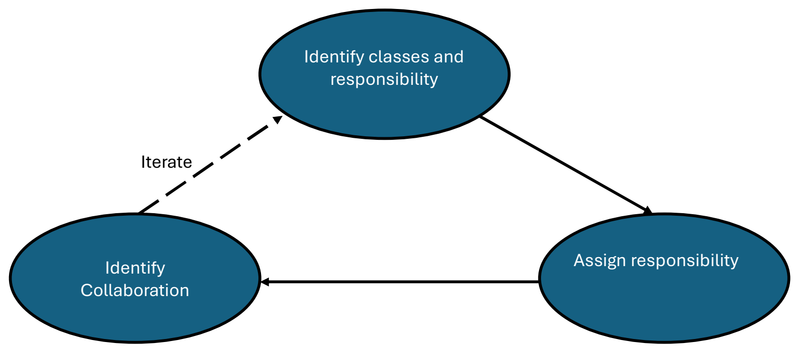
CLASSES, RESPONSIBILITIES, AND COLLABORATORS (CRC)

A CRC index card



CLASSES, RESPONSIBILITIES, AND COLLABORATORS (CRC)

The CRC Process



CLASSES, RESPONSIBILITIES, AND COLLABORATORS (CRC)

CRC for the Account Object

Account	CheckingAccount
balance	(subclass)
Numbers	Savings Account (subclass)
-------------	Transaction
Deposit	
Withdraw	
getBalance	

CLASSES, RESPONSIBILITIES, AND COLLABORATORS (CRC)

Naming Classes-Guidelines

✓The class name should be singular

✓Use names which are comfortable for users.

✓The name of the class should reflect its intrinsic nature.

✓Use readable names.
- By convention, the class name must begin with an upper-case latter.
- For compound words, capitalize the first letter of each word- LoanWindow

Further Reading

Chapter07 from Ali Bahrami textbook